

Industrial Composition and Marriage Timing in China's Third Front*

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Abstract

Did industrialization under central planning leave a distinctive marriage-age profile? We study China's Third Front with the 1982, 1990, and 2000 population censuses. In 1982, a one-standard-deviation increase in the heavy-industry share is associated with a 4.23-percentage-point larger gap in never marriage between ages 20–24 and 30–34 inside Third Front provinces than elsewhere; predetermined siting controls reduce the differential to 2.24 points. Retrospective first-marriage histories observed in 2000 yield similar pseudo-1982 point estimates under birth-year and completed-age indexing. The point-estimate alignment survives on identical respondents, although uncertainty permits meaningful differences and current residence cannot recover the 1982 population. Campaign-aligned comparisons show an early marriage deficit and subsequent catch-up within industrial Third Front areas, but no robust additional shift relative to other areas. Education shares account for 15.5 percent of a comparable young–old gap. The paper establishes a persistent resident-population marriage-age gradient and a transparent bridge from census stocks to cohort histories, while showing why neither identifies the campaign's causal effect.

Keywords: marriage timing; industrialization; work units; place-based policy; Third Front; China.

JEL codes: J12, J23, N35, O25, P23.

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1 Introduction

Industrialization can reshape family formation through more than wages and employment. Under central planning, factories also assigned jobs, rationed housing, regulated residence, and supplied documents needed to form a household. China’s Third Front—a two-decade campaign that placed defense and heavy industry in the interior—therefore offers a setting in which to ask whether administratively industrialized places developed a distinctive marriage-age profile, and what repeated censuses can reveal about its source.

The main finding is a pronounced young-singlehood gradient in industrial prefectures inside Third Front provinces. In the 1982 census, a one-standard-deviation increase in the heavy-industry share is associated with a 4.23 percentage points larger difference in never-married rates between ages 20–24 and 30–34 inside designated provinces than elsewhere. Predetermined siting controls reduce the estimate to 2.24 points. Retrospective marriage histories observed in 2000 yield point estimates of 3.95 and 3.86 points under birth-year and completed-age indexing. Yet campaign-aligned cohorts show no robust additional Third Front shift relative to other areas. The pattern is persistent; its attribution is not automatic.

Three questions organize the paper. First, did the population living in industrial places in 1982 display a distinctive resident-age gradient? Second, does that gradient reappear when marital status is reconstructed for the corresponding cohorts? Third, did cohorts whose marriage risk occurred during the campaign change differently from cohorts whose risk occurred before it? The answers are, respectively, yes, approximately yes in point estimates, and no robust Third-Front-specific difference in the available current-residence design. Keeping these questions separate is the paper’s organizing contribution.

We use the one-percent microdata from the 1982, 1990, and 2000 population censuses ([Minnesota Population Center, 2020](#)). A province-validated crosswalk holds fixed 148 prefectures in 21 provinces across waves. Industrial composition is measured by the 1980 heavy-industry share reconstructed from 253 million firm-registration records, with plant employment, investment, and designated industrial centers from [Fan and Zou \(2021\)](#) used as sharper alternatives. The core design compares never-married rates at ages 20–24 and 25–29 with rates at 30–34 within prefecture, and asks whether that age gradient changes more steeply with industrial composition inside Third Front provinces than outside.

The resident evidence is economically large but geographically sensitive. The baseline 1982 differential is 4.23 points per standard deviation; the slope within designated provinces is 4.82 points, compared with 0.59 elsewhere. Predetermined siting controls reduce the differential by roughly half, and the unweighted estimate is less precise. We therefore interpret 2.2–4.2 points as a descriptive range rather than select one invariant effect. The age profile recurs on the same geography in 1990 and 2000, which establishes a repeated resident-population pattern rather than a persistent cohort effect.

The retrospective evidence answers the measurement question. We use reported first-marriage histories in 2000 to reconstruct pseudo-1982 marital status for the reported 1948–1962 birth years. The birth-year and matched completed-age differentials are 3.95 and 3.86 points. On the 1,316,216 respondents common to both age-index supports, changing the index moves the estimate by only -0.08 points (wild $p = 0.341$). In joint models, the birth-year profile differs from the actual 1982 resident stock by -0.28 points (wild $p = 0.797$), with a 95-percent interval of $[-2.45, 2.20]$. Thus the point estimates align after transparent age-index translation, while uncertainty and different resident populations leave room for meaningful differences.

The campaign-cohort comparison answers the timing question. The original 1982 age groups passed through marriage risk during overlapping parts of the 1964–1980 build-out and therefore cannot isolate a campaign effect. When we compare marriage-risk windows that lie wholly before the campaign or within the build-out, industrial Third Front areas show an early deficit that peaks near age 23 and catches up by 25. The pattern is not family-wise robust across thresholds, and the additional Third-Front-versus-other shift is imprecise throughout. Current residence in 2000 also remains an imperfect assignment of historical exposure.

The remaining evidence narrows several simple accounts without selecting a mechanism. An exact decomposition assigns 0.59 of 3.80 points, or 15.5 percent, to education shares and 3.21 points to marriage-rate differences within education groups (Kitagawa, 1955); the latter component is imprecisely estimated under wild inference. Student status, collective residence, and aggregate sex ratios do not explain the profile. Public-housing dependence in the 1988 urban household survey and historical evidence on isolated factory compounds make a work-unit channel plausible (Marukawa, 2021), but no available dataset links housing allocation to individual marriage timing. Direct plant measures, the railway design, and the CFPS associations remain secondary evidence.

This paper makes two contributions. First, it documents a previously unreported resident-age gradient in industrial Third Front places and shows that later first-marriage histories retain a similar point profile. This adds an institutional setting to work on labor demand and family formation, which typically studies market-mediated employment shocks (Autor, Dorn and Hanson, 2019; Braga, 2018), and to research showing how migration and education reshape Chinese marriage markets (Angrist, 2002; Dupuy, 2021; Xiong, 2023; Mu and Yeung, 2020; Fang et al., 2026; Rossi and Xiao, 2026). Second, it shows how repeated censuses can separate measurement alignment from treatment attribution in historical place-based research. Age-specific proportions never married are population stocks (Hajnal, 1953); cohort interpretation requires recoverable membership (Deaton, 1985; Ryder, 1965), and selective mobility can separate repeated cross sections from longitudinal histories (Lubotsky, 2007). The distinction matters whenever policy, migration, and family formation jointly determine who is observed in a place (Kline and Moretti, 2014; Busso, Gregory and Kline, 2013; Criscuolo et al., 2019; Glaeser and Gottlieb, 2008; Kline and Moretti, 2013; Gerritse, Wang and van Oort, 2026).

Within the Third Front literature, [Fan and Zou \(2021\)](#) study industrial siting and local development, while [Fang, Liang and Qin \(2026\)](#) study human-capital consequences with family outcomes downstream. The closest concurrent project uses factory timing to study marriage among exposed women.¹ Our analysis complements that design by asking whether a current-residence industrial gradient survives translation from a census stock into a retrospective cohort profile. [Tian \(2016\)](#) similarly uses repeated Chinese censuses to trace synthetic-cohort transitions, while [Song and Zheng \(2016\)](#) illustrates how historical residence measured before outcomes can support a treatment interpretation.

The paper proceeds from institutional setting and data to the empirical framework, the resident and retrospective evidence, and campaign-cohort interpretation. The registry share is an endogenous measure of industrial composition, current residence is not birthplace, and the available project files contain no activation dates; these limits determine the scope of the claims below.

2 Institutional Setting and Data

2.1 The Third Front and work units

The Third Front was launched in 1964 after Chinese leaders came to view the coastal industrial base as vulnerable to attack. The campaign moved defense and heavy industry into the Southwest, Northwest, and interior Central China, building rail, power, and urban infrastructure around it. Siting followed strategic and topographic criteria—dispersed, concealed, often mountainous locations away from existing cities—and investment slowed in the late 1970s, though many plants remained major employers afterward. For identification these features pull in opposite directions: siting responded less to market demand than a typical industrial expansion, but projects were not randomly placed within the interior; rail access, terrain, existing capacity, and administrative priorities all mattered. [Fan and Zou \(2021\)](#) address placement with distance to the pre-1962 railway network, conditional on the completed 1980 network and geography; we use their variables as a demanding robustness exercise without inheriting their exclusion restriction for marriage.

Urban industrial employment was organized through the *danwei*. A work unit was employer, welfare provider, and gatekeeper at once: it assigned work, administered wages, allocated housing, and—until the 2003 regulations—supplied the marital-status documentation required to register a marriage ([Davis, 2014](#)). The bundling admits mechanisms of both signs. Housing queues and seniority rules could delay marriage; stable jobs and subsidized housing could ease it; isolated compounds could narrow partner pools; administrative assignment could alter education and its opportunity cost of early marriage. The historical record makes isolation concrete: in Shanghai’s Small Third Front in southern Anhui, factories imported inputs and employees from Shanghai

¹Public materials describe county-by-factory variation from roughly 600 plants ([Gao, Zhang and Zhou, n.d](#); [School of Labor Economics, Capital University of Economics and Business, 2025](#)). No manuscript, data, or replication package was publicly available to us as of 30 August 2026; we therefore cite only the public description. Its factory-timing design and our current-residence prefecture design answer different questions and can yield different results.

and were socially separate from surrounding villages, with most employees marrying within the industrial community (Marukawa, 2021). A queue account carries a specific timing implication—more singlehood at young ages *and* later catch-up within the same cohorts—and observing only the first, cross-sectionally, is insufficient. The work-unit system also outlived the construction campaign: job assignment, public housing, and residence controls remained central through the 1980s and unwound gradually in the 1990s, which is why repeated censuses and the 2000 retrospective histories provide informative boundaries.

2.2 Data and measurement

Outcomes come from the one-percent samples of the 1982, 1990, and 2000 population censuses distributed by IPUMS International (Minnesota Population Center, 2020). The cross-sectional profiles retain ages 18–40 with valid age, sex, marital status, weight, and geography and aggregate to prefecture-by-single-age-by-sex cells with person weights. The outcome is an indicator for never married. The microdata also identify current students, collective dwellings, and education strata; the 1990 census records residence in the same minor administrative unit five years earlier, and the 2000 census the same major unit, a province-level stability measure we label literally. The 2000 census additionally reports age and year at first marriage for the ever married; combined with current status, this yields marriage-by-each-age outcomes among older respondents who were unmarried at twenty, with the two constructions agreeing within one year for 99.995 percent of ever-married respondents. The broad history analysis covers 3.22 million people; the 1948–1962 birth-year bridge maps 1.34 million 2000 respondents to the common geography. These data support synthetic-cohort descriptions (Tian, 2016), but current prefecture is not historical residence.

Because prefecture names are not unique across China, the census-to-prefecture crosswalk parses the province from each IPUMS geographic code and requires it to match the first two digits of the 1982 prefecture code; name-only matching would admit 13 false candidates in 1982 and 10 in 1990. The province-constrained match yields 188 prefectures in 1982 and 233 in 1990; intersecting the 1982–1990–2000 supports with a valid exposure leaves 154, and dropping provinces contributing a single prefecture leaves 148 in 21 provinces. All cross-census comparisons hold this geography fixed, and a validation script enforces crosswalk uniqueness, sample counts, and manifest freshness (Appendix A).

Broad exposure is the heavy-industry share—manufacturing plus mining—of cumulative registered entries in a prefecture through 1980, constructed from 253 million raw firm-registration records with founding years in an author-licensed MacroData extract downloaded in August 2026 (MacroData, 2026). The registration system was rebuilt around the reform era, so pre-1980 records are retrospective founding dates rather than a contemporaneous plant census, and measurement error need not be classical; the variable is best understood as a noisy record of inherited industrial composition. Exposure distributions overlap extensively across designation status (means 0.500 and 0.469; Appendix Figure B.1). The overlap reduces extrapolation but validates neither linearity nor

exogeneity. As sharper alternatives we use the [Fan and Zou \(2021\)](#) prefecture files: 1985 large- and medium-plant employment relative to 1982 employment, Third Front investment, the ten designated industrial centers, rail distances, and siting controls. Designation follows the historical program at the province level (Yunnan, Guizhou, Sichuan including Chongqing, Shaanxi, Gansu, Ningxia, Qinghai, Shanxi, Henan, Hubei, Hunan) and is absorbed by prefecture fixed effects. Qinghai and Ningxia contribute only singleton prefectures and drop from estimation, leaving nine designated provinces on the 148-prefecture support.

Two auxiliary sources measure institutions and provide an independent survey design. The 1988 urban China Household Income Project (CHIP) records whether the household head worked in industry and whether the dwelling was publicly owned, for 8,094 households in ten provinces, five designated ([Griffin and Zhao, 2010](#)). The 2010–2022 China Family Panel Studies (CFPS) waves provide retrospective first-marriage histories ([Xie and Hu, 2014](#); [Institute of Social Science Survey, Peking University, 2015](#)); because the access-controlled project extract identifies survey-era city rather than pre-marriage location, its estimates are secondary (Section 5.2). City and respondent identifiers are not redistributable and are excluded from the public replication package.

3 Empirical Framework

The analysis separates three empirical objects: a resident-age gradient observed at a date, a retrospective cohort profile reconstructed from later marriage histories, and a campaign-cohort comparison defined by marriage-risk windows. Appendix Table C.1 records the population and contrast behind each object. The separation matters because the same-looking age profile can combine marriage timing and resident composition.

Formally, if g indexes groups defined by cohort and observed or unobserved characteristics, the resident stock at age a can be written as

$$R_{pa} = \sum_g \Pr(g \mid p, a, \mathcal{R}) [1 - F_{gp}^{\mathcal{R}}(a)], \quad (1)$$

where $\mathcal{R}$ denotes selection into the observed resident population and $F_{gp}^{\mathcal{R}}(a)$ is the first-marriage distribution within that selected population. An industrial gradient in R_{pa} can therefore arise from group-specific marriage timing, from the composition weights, or from both.

3.1 Resident-age gradient

Let N_{pas} be the weighted never-married share in prefecture p , age a , sex s ; let H_p be the 1980 heavy share standardized by its cross-prefecture standard deviation (0.137, computed on the 154-prefecture

common support before dropping singleton provinces), and T_p Third Front designation. Estimation uses 148 prefectures in 21 provinces. The core specification is

$$N_{pas} = \alpha_p + \lambda_{q(p)as} + \sum_{g \in \{20-24, 25-29\}} [\beta_g H_p \mathbf{1}\{a \in g\} + \delta_g H_p T_p \mathbf{1}\{a \in g\}] + \varepsilon_{pas}, \quad (2)$$

with ages 30–34 omitted, prefecture fixed effects α_p , and province-by-single-age-by-sex effects $\lambda_{q(p)as}$; cells are weighted by population and clustered by province, and stacked versions add census interactions. The headline coefficient δ_{20-24} asks whether the relationship between industrial composition and young singlehood bends more sharply at young ages inside designated provinces, relative to the same prefectures at 30–34. Prefecture effects absorb permanent local attributes, including designation itself; the saturated province-age-sex effects absorb every province-specific marriage-age profile. What survives as a threat is any prefecture characteristic that covaries with H_p and tilts the age profile differentially inside designated provinces—education is the leading example, and the results treat it as such. With 21 clusters, analytic p -values are supplemented by restricted-null wild cluster bootstraps (4,999 draws) and by exact reassignment of the nine designation labels among the 13 interior provinces; historical assignment was not uniform over that set, so the exact tail areas are design diagnostics, not randomization inference.

3.2 Retrospective histories

The age-index bridge asks a measurement question. For each 2000 respondent with reported birth year 1948–1962, the primary index is $a^{BY} = 1982 - \text{birth year}$; we reconstruct whether first marriage had occurred by a^{BY} . Actual 1982 cells instead use completed age, and birth months are unavailable, so this is a near alignment rather than exact linkage. The completed-age sensitivity uses $a^{CA} = \text{recorded completed age in 2000 minus 18}$. The unrestricted age-20–34 boundary samples under a^{BY} and a^{CA} contain 1,339,802 and 1,327,590 respondents, with 1,316,216 in both. We re-estimate both indices on those identical respondents so their direct difference isolates indexing rather than boundary composition. Within the full birth-year sample, the two indices differ by one year for 17.1 percent of records; different census months mean that neither exactly recovers completed age on the 1982 census date. We re-estimate Equation (2) on each pseudo-1982 profile using residence in 2000 and stack actual and pseudo cells with source-specific prefecture and source-specific province-by-age-by-sex effects. Pseudo-minus-actual interactions, their covariance, and wild tests are therefore estimated jointly over the same 21 provinces. Age at first marriage is primary; marriage year provides a third calendar-age sensitivity.

3.3 Campaign cohorts and inference

The campaign comparison asks a treatment-timing question and reports three slopes. Cohorts born 1944–1960 reached age twenty during the build-out, while cohorts born 1934–1943 did so before it. For marriage by each age $a = 21, \dots, 28$ among people unmarried at twenty, a model with

prefecture and province-by-single-birth-year-by-sex effects reports the slope shift outside designated provinces, the additional Third-Front differential, and their sum—the net shift inside designated provinces. We use restricted-null wild inference for both the differential and the net shift and adjust each eight-threshold family by Holm’s method. Because these broad cohort labels still permit marriage-risk windows to straddle 1964, a stricter diagnostic changes the cohorts at each threshold: the full ages-21-to- a risk window must end by 1963 or lie wholly within 1964–1980, and the nearest pre-program window contains the same number of birth cohorts as the build-out window. This removes calendar overlap at the cost of older, survivor-selected comparison cohorts. Every history design assigns exposure by residence in 2000, not by birthplace or residence when marriage risk began.

4 Resident-Stock and Retrospective Evidence

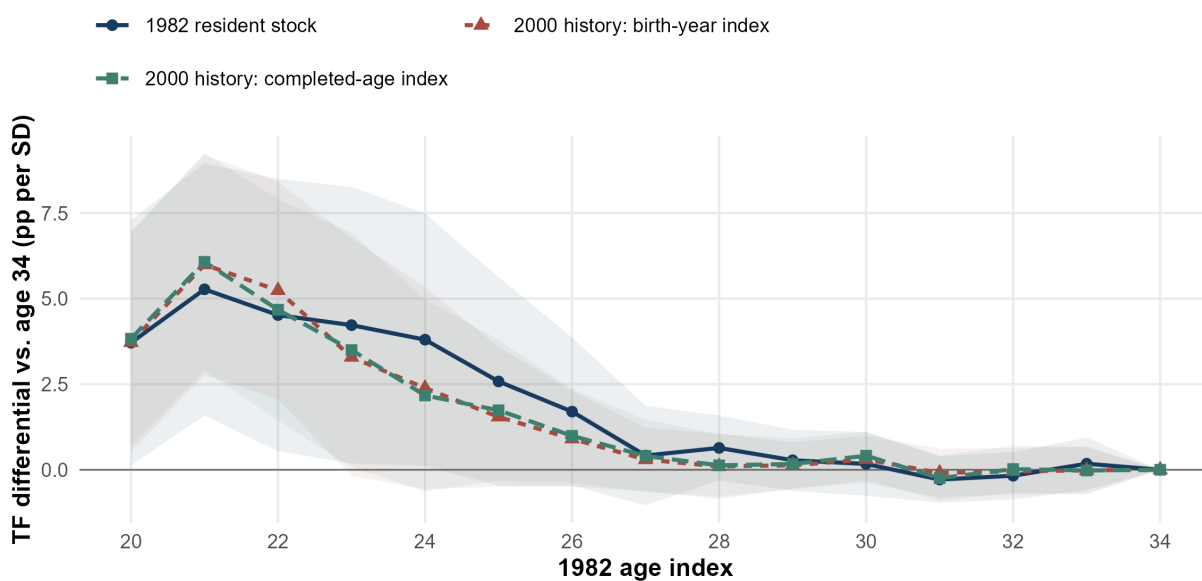
4.1 Resident-age gradient

Figure 1 shows the paper’s central empirical result: the young–old singlehood gradient among 1982 residents reappears when marital status is reconstructed from the 2000 census histories. With age 34 as the reference, the 1982 differential peaks near 5.3 percentage points per standard deviation at age 21, remains elevated through 24, and approaches zero around age 30. The two retrospective profiles follow the same broad age pattern despite different survival and residence selection.

Table 1 contains the grouped estimates and the full inference. The 1982 baseline differential is 4.23 points (s.e. 1.66) at 20–24 and 1.14 (0.66) at 25–29; the net slope inside designated provinces is 4.82 points against 0.59 outside, so the differential is primarily a young-age gradient of the designated interior. Wild-bootstrap inference over the 21 provinces gives $p = 0.039$ at 20–24 and $p = 0.105$ at 25–29, and no single province drives the estimate: leave-one-province-out estimates span 3.54 to 5.12 points with 20 of 21 intervals excluding zero (Appendix Figure B.2). The diagnostics do not reject linearity (median-knot spline $p = 0.67$), but the nonmonotone quartile pattern cautions against treating the slope as a structural dose response: the top-versus-bottom contrast is 8.99 points ($p = 0.039$), while the quartile coefficients are not jointly distinguishable ($p = 0.19$; Appendix Table B.1).

Predetermined siting controls reduce the headline gradient by roughly half. Interacting pre-campaign investment, 1962 rail distance, terrain, elevation, distance to the provincial capital, 1964 urbanization and density, and 1936 employment with the age bins lowers the differential to 2.24 points (s.e. 1.15; wild $p = 0.099$); the unweighted estimate is 2.51 ($p = 0.128$). The completed 1980 rail network is post-campaign and therefore excluded from this control set. Under exact designation reassignment the controlled estimate’s tail area is 0.249, against 0.059 for the baseline. Geographic sensitivity is therefore part of the substantive result: we interpret 2.2–4.2 points as a descriptive range rather than as a stable treatment effect.

Figure 1. Resident stocks and retrospective histories under alternative age indices



Notes. The actual series uses never-married stocks among 1982 residents. Retrospective series reconstruct pseudo-1982 status from age at first marriage among 2000 survivors/residents, indexing age either as 1982 minus reported birth year or as 2000 completed age minus 18. The completed-age series uses the 1,316,216 respondents whose two indices both lie from 20 to 34; the full birth-year series retains all 1,339,802 reported 1948–1962 births. Within that full sample, the indices differ by one year for 17.1 percent of records. Coefficients are age-specific heavy-share-by-Third-Front differentials relative to age 34, scaled by the common-support standard deviation. Models contain prefecture and province-by-age-by-sex effects, use population weights, and cluster by province. Pseudo profiles assign industrial composition by residence in 2000 and do not reconstruct the 1982 resident population.

Table 1. The Third Front differential in the census marriage-age profile

	Ages 20–24	Ages 25–29	Slope within TF, 20–24
<i>Panel A. 1982 census</i>			
Baseline	+4.23** (1.66)	+1.14 (0.66)	+4.82*** (0.86)
Predetermined siting controls	+2.24* (1.15)	+0.18 (0.49)	+2.16*** (0.74)
Siting controls, unweighted	+2.51 (1.58)	—	—
Nonstudents, household dwellings	+4.38** (1.69)	+0.91 (0.62)	+4.91*** (0.87)
Education-stratum fixed effects	+2.27 (1.43)	+0.86 (0.52)	+2.69*** (0.66)
<i>Panel B. Later censuses, same geography</i>			
1990 baseline	+2.33* (1.26)	+0.31 (0.62)	+2.60** (0.92)
1990, same minor unit 1985	+2.27* (1.31)	+0.42 (0.58)	+2.50** (0.97)
2000 baseline	+3.15*** (0.73)	+1.42*** (0.46)	+3.07*** (0.62)
2000, nonstudent, same major unit 1995	+2.74*** (0.84)	+1.28*** (0.42)	+2.78*** (0.81)
2000, education-stratum fixed effects	+2.28*** (0.64)	+1.11*** (0.34)	+1.83*** (0.54)
Difference from 1982, 1990 census	−1.90* (0.97)	−0.83* (0.43)	—
Difference from 1982, 2000 census	−1.08 (1.51)	+0.28 (0.49)	—

Notes. Equation (2) on the common 148-prefecture, 21-province geography; cells weighted by population; standard errors clustered by province. Entries are percentage points per cross-prefecture standard deviation (0.137) of the 1980 heavy-industry share; age bins are contrasts against ages 30–34 within prefecture. The net Third Front slope is the sum of the ordinary and interaction slopes. Restricted-null wild cluster bootstrap over the 21 provinces (4,999 draws): baseline 1982 differential $p = 0.039$, siting-control differential $p = 0.099$. Exact reassignment of the nine designation labels among the 13 interior provinces: baseline $p = 0.059$, siting-controlled $p = 0.249$. Leave-one-province-out baseline estimates span 3.54 to 5.12 points, with 20 of 21 conventional 95-percent intervals excluding zero. Siting controls interact pre-campaign investment, 1962 rail distance, terrain, elevation, distance to the provincial capital, 1964 urbanization and density, and 1936 employment with the age bins. The completed 1980 rail network is post-campaign and deliberately excluded. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Panel B of Table 1 shows recurrence on the identical geography. The 20–24 differential is 2.33 points in 1990 and 3.15 in 2000; a stacked model estimates the 2000-minus-1982 change at −1.08 ($p = 0.48$), while the 1990-minus-1982 dip is −1.90 (wild $p = 0.114$). These repetitions establish a recurring resident-age profile, not a persistent cohort effect: age and cohort determine census year, so the repeated cross sections cannot separately identify all three. The 2000 restriction to nonstudents in the same major unit five years earlier gives 2.74 points, and the 1990 same-minor-unit subsample gives 2.27 (s.e. 1.31); these five-year restrictions do not address lifetime sorting.

4.2 Retrospective age-index bridge

Table 2 aggregates the profiles in Figure 1.

Table 2. Age-index bridge between the 1982 resident stock and retrospective histories

	Ages 20–24 versus 30–34			Ages 25–29 versus 30–34		
	Estimate	(s.e.)	Wild p	Estimate	(s.e.)	Wild p
<i>Panel A. Third Front differential</i>						
1982 resident stock	+4.23	(1.66)	0.039	+1.14	(0.66)	0.105
2000 history: birth-year index	+3.95	(1.37)	0.015	+0.56	(0.41)	0.205
2000 history: completed-age index, matched	+3.86	(1.36)	0.014	+0.65	(0.42)	0.150
2000 history: marriage-year sensitivity	+3.75	(1.42)	0.025	+0.72	(0.49)	0.185
<i>Panel B. History minus 1982 resident stock</i>						
Birth-year index, marriage age	−0.28	(1.01)	0.797	−0.58	(0.39)	0.145
Completed-age index, matched records	−0.37	(1.00)	0.714	−0.49	(0.39)	0.205
Birth-year index, marriage year	−0.48	(1.00)	0.639	−0.41	(0.37)	0.272
<i>Panel C. Index difference on the same respondents</i>						
Completed-age minus birth-year index	−0.08	(0.08)	0.341	+0.09	(0.05)	0.114

Notes. Entries are percentage points per cross-prefecture standard deviation (0.137) of the 1980 heavy-industry share. All models use the common 148-prefecture, 21-province support, prefecture fixed effects, province-by-single-year-age-by-sex effects, population weights, and province clustering. The actual estimand is the never-married stock among 1982 residents. The primary pseudo profile uses the exact reported 1948–1962 birth cohorts and indexes age as 1982 minus birth year. The completed-age sensitivity uses the 1,316,216 respondents whose birth-year and completed-age indices both lie from 20 to 34. Panel C re-estimates both indices on those identical respondents, isolating the index change from boundary composition. Within the full birth-year sample, 17.1 percent differ by one year under the two indices. Neither observes birth month or exact completed age on the 1982 census date. Both use 2000 first-marriage histories, assign exposure by residence in 2000, and condition on survival and residence in China through 2000. Wild p -values use 4,999 restricted-null Rademacher draws over provinces.

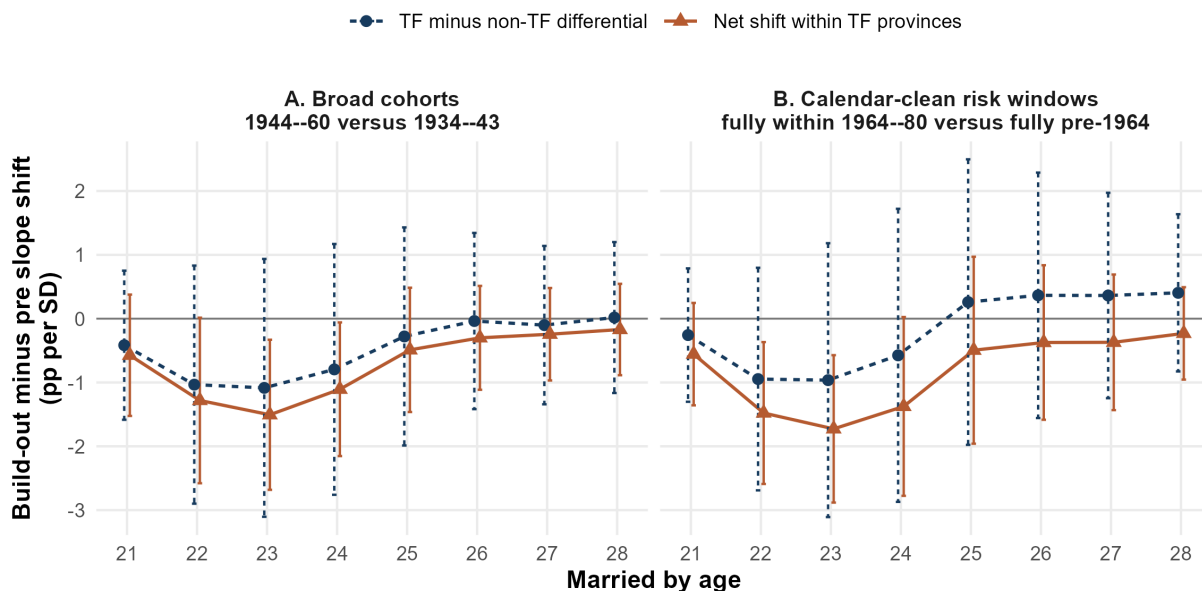
The actual, full-sample birth-year, and matched-sample completed-age 20–24 differentials are 4.23, 3.95, and 3.86 points. Re-estimating the birth-year index on the same matched records gives 3.95 points, and the direct completed-minus-birth-year difference is −0.08 points (s.e. 0.08; wild $p = 0.341$). In joint stacked models, the two pseudo-minus-actual estimates are −0.28 points (s.e. 1.01; wild $p = 0.797$; wild interval $[-2.45, +2.20]$) and −0.37 points (s.e. 1.00; wild $p = 0.714$; interval $[-2.41, +2.11]$). Reconstructing timing from marriage year gives 3.75 points and a difference of −0.48 (wild $p = 0.639$). This comparison answers a measurement question, not a treatment question. It shows that the age pattern is not created solely by the choice of age index, but it cannot rule out sorting or marriage-related migration because the histories assign industrial composition by residence in 2000.

5 Campaign Cohorts and Interpretation

5.1 Campaign-cohort comparisons

The aligned retrospective profile asks whether the original age pattern is visible for the corresponding reported birth years. Figure 2 and Appendix Table C.2 ask the separate timing question: how the heavy-industry slope changes from pre-program to build-out cohorts outside designated provinces, how much more it changes inside Third Front provinces, and the resulting net change within those provinces. A negative estimate means lower marriage completion for build-out cohorts.

Figure 2. Broad and calendar-clean cohort timing across fixed-age thresholds



Notes. The broad comparison uses births 1944–1960 versus 1934–1943. The calendar-clean comparison changes the birth windows at each threshold so the full ages-21-to-threshold risk period falls either before 1964 or within 1964–1980; the two windows contain equal numbers of birth cohorts and sit next to the policy boundary. Outcomes are marriage by the indicated age among people unmarried at 20. Models include prefecture fixed effects and province-by-single-birth-year-by-sex effects, weight by population, and cluster by province. Error bars are conventional 95-percent intervals; Table C.2 reports restricted-null wild and Holm-adjusted p -values. Industrial composition is assigned from prefecture of residence in 2000.

The broad comparison reveals an early deficit followed by catch-up. The net slope within designated provinces reaches -1.51 points by age 23 and narrows to -0.49 by 25 and -0.17 by 28. The age-23 estimate has a pointwise restricted-null wild $p = 0.039$, but its Holm family-wise p across eight thresholds is 0.315. The additional Third-Front differential ranges from -1.08 to $+0.02$ points, and every Holm-adjusted p -value is one. The data therefore show a short-delay pattern within industrial Third Front areas, not a distinct family-wise-robust shift relative to other areas.

The non-overlapping risk windows sharpen rather than overturn that reading. Their net estimate reaches -1.73 points by age 23 and narrows to -0.49 by 25. The raw wild p -values at ages 22 and 23 are 0.015 and 0.027, but become 0.118 and 0.188 after Holm adjustment; the corresponding

Third-Front differentials never approach family-wise significance. The birth-year diagnostic in Appendix Figure C.1 likewise shows no discrete differential break at the campaign boundary.

The uncertainty and assignment rule set the interpretation. A Holm correction does not establish equivalence or rule out a modest, short-lived effect; the calendar-clean comparison uses older, survivor-selected pre-program cohorts; and all histories assign industrial composition by residence in 2000. A factory-level effect on people observed where they lived before exposure—including the design publicly described by Gao, Zhang, and Zhou—could therefore escape this comparison.

5.2 Composition and alternative explanations

Education composition. The birth-year-indexed similarity does not rule out sorting, so we next ask whether specific observed composition margins account for the pattern. Secondary-schooling shares have a 4.33-point young–old differential in the core 1982 design (Appendix Table B.2). Merely adding education fixed effects reduces the marriage differential from 4.23 to 2.27 points, but coefficient attenuation across differently weighted fixed-effect specifications is not a decomposition. Table 3 instead applies the exact symmetric decomposition of Kitagawa (1955) to prefecture-by-sex young–old gaps. With two exhaustive education groups, the 3.80-point collapsed-gap differential separates exactly into 0.59 points from education shares and 3.21 points from marriage-rate differences within education. In this point decomposition education composition is 15.5 percent, not an explanation for half of the profile and not a causal mediator. The 84.5-percent within-education point component remains imprecise under wild inference ($p = 0.118$); it may contain institutional timing, unobserved composition within broad schooling groups, or both.

Table 3. Education composition and the young–old marriage gap

Outcome	Differential (SE)	Wild p	Within-TF (SE)	Wild p
Observed young-old gap	3.80 (1.81)	0.078	4.38 (0.86)	0.005
Education-composition component	0.59 (0.18)	< 0.001	0.70 (0.15)	0.004
Common-weight within-education component	3.21 (1.74)	0.118	3.68 (0.81)	0.005

Notes. Differential entries are the heavy-industry-share slope inside Third Front provinces minus the slope elsewhere; within-TF entries are the net slopes inside Third Front provinces. Both are per a one-standard-deviation increase in the 1980 heavy-industry share ($SD = 0.137$). Outcomes are percentage points of the ages 20–24 minus ages 30–34 never-married gap. Education is below-secondary versus secondary-plus. The composition and common-weight within-education components add exactly to the observed gap. Regressions contain province-by-sex fixed effects, use harmonic-mean population weights, cluster by province, and use 4,999-draw restricted-null wild inference.

Other composition margins. Removing students and collective dwellings leaves the main marriage differential at 4.38 points, and the profile appears for both women (4.67 points) and men (3.79). On the audited 148-prefecture support, the male share among never-married residents ages 20–34 has a negative but imprecise -1.47 -point differential (s.e. 1.14; wild $p = 0.235$), providing

little support for a simple male-inflow squeeze. Appendix Table D.1 reports the full sex-composition diagnostic. Five-year stayer restrictions also leave the profile near baseline, but observe neither childhood location nor moves around first marriage.

Institutional evidence. In the 1988 CHIP urban sample, industrial households were *less* dependent on public housing than their neighbors outside designated provinces (78.4 versus 85.2 percent) and *more* dependent inside them (90.2 versus 87.0). The adjusted 11.0-point difference-in-differences has a ten-cluster wild-bootstrap $p = 0.017$ and an exact reallocation $p = 0.012$ (Appendix Table D.4). This pattern makes a work-unit housing channel plausible in the institution’s own era, but CHIP records no waiting list, allocation date, rejected certificate, or marriage outcome.

Secondary designs. The CFPS yields Third-Front-versus-other slope differentials of -1.7 points by 25 and -2.0 by 28, but the imprecisely estimated net slopes inside designated cities are only about -0.3 and -0.6 ; location is the survey-era city and the differential weakens under province-by-cohort effects (Appendix D.4). Project-capacity associations are sensitive to support, direct census investment and capacity profiles fall from 1.61 and 2.34 points to 0.46 and 0.92 under siting controls, and the designated-center coefficients are descriptive with only seven effective provinces. The railway IV’s conventional estimate is positive, but its weak-instrument-robust wild Anderson–Rubin p -value is 0.366 on a 22-prefecture overlap (Appendix Tables D.2 and D.3). These designs remain secondary because none assigns industrial exposure before migration with adequate timing and support.

Taken together, the channel checks narrow several simple explanations without identifying a mechanism. Education shares account for a minority of the comparable gap, while student status, collective residence, and aggregate sex composition provide little support for their corresponding accounts. Public-housing dependence and within-compound matching make a work-unit channel plausible, but neither is linked to individual waiting time or marriage. Lifetime sorting and marriage-related migration remain possible because historical residence is unobserved.

5.3 Identification and data requirements

The evidence establishes a resident-age gradient of 2.24–4.23 points on the maintained linear industrial-composition scale, a similar point profile in reconstructed first-marriage histories, and a measured minority role for education shares in a comparable gap. These findings jointly describe the demographic imprint of industrial places. They do not estimate an average treatment effect: the registry share is endogenous, actual and pseudo profiles select residents at different dates, current residence can respond to marriage, and the project files contain no activation dates. The weak campaign-cohort differential likewise does not establish equivalence or rule out factory-level effects assigned at birthplace or pre-marriage residence.

A causal design requires a county-level roster of project approval, construction, and production dates linked to historical residence. Such data would permit cohort-by-county comparisons with

pretrends and placement-appropriate inference. Restricted census birthplace codes would assign exposure before migration, while personnel and housing-allocation archives could test directly whether work-unit queues delayed household formation.

6 Conclusion

Industrialized prefectures inside Third Front provinces display a markedly steeper young-singlehood profile than comparable places elsewhere. Depending on how much predetermined siting geography is held fixed, the difference in never-married rates between ages 20–24 and 30–34 is 2.2–4.2 percentage points larger per standard deviation of industrial composition. Retrospective first-marriage histories observed in 2000 retain a similar point pattern, and measured education composition explains only a minority of a comparable gap.

These findings establish a persistent demographic regularity, but they do not identify the Third Front’s effect on marriage. The resident stock and retrospective histories select people at different dates, current residence may reflect prior marriage and migration, and the original age groups passed through overlapping campaign years. The paper’s contribution is to show that aligning age definitions can bring census stocks and cohort profiles into the same empirical frame while leaving the treatment question unresolved.

A causal answer requires project activation dates linked to birthplace or pre-marriage residence. County-level project histories would assign campaign exposure before migration, while personnel and housing-allocation archives could test directly whether work-unit queues delayed household formation.

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Online Appendix

Industrial Composition and Marriage Timing in China's Third Front

Yi Yao Tai Lo Yeung

1 September 2026

Appendix contents

A. Data construction and crosswalk validation	A
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A Data Construction and Crosswalk Validation

The census files identify geographic units with survey-specific numeric codes and English value labels; the Fan–Zou data identify 1982 prefectures with four-digit codes and names. We normalize labels by lowercasing, removing administrative suffixes, and collapsing whitespace, and permit a match only when the first two digits of the prefecture code equal the census province code; combined geographic units in the IPUMS labels are excluded, and each census code must map to at most one prefecture. Name-only matching would admit 13 false candidate matches in 1982 and 10 in 1990—same-named units in different provinces—which the province constraint removes. The constrained match leaves 188 matched 1982 prefectures (157 with a valid heavy share) and 233 in 1990 (189 valid); intersecting the 1982–1990–2000 supports leaves 154, and dropping single-prefecture provinces leaves 148 in 21 provinces. The validation suite enforces crosswalk uniqueness, fixed support, sample counts, artifact freshness, and the reported numerical identities; implementation details appear in `REPLICATION.md`.

B Resident-Stock Diagnostics and Robustness

B.1 Cross-sectional age-bin slopes and grouped estimates

The DDD of Equation (2) is built from three cross-sectional slopes. In the common 1982 geography, the Third Front differential in the 20–24 heavy-share slope is 4.21 points per standard deviation when ages are estimated separately, against 1.06 at 25–29 and approximately zero at 30–34; removing students and collective dwellings gives 4.38, 0.89, and zero; education adjustment gives 3.09, 1.38, and 0.34. In 1990 the young cross-sectional differential is 2.10 points and in 2000 it is 3.76; the same-major-unit 2000 restriction gives 2.99. These estimates do not absorb prefecture levels, which is why the body emphasizes Equation (2). Figure B.3 displays the grouped DDD estimates on the identical geography across census years.

Table B.1. Functional-form and province-designation diagnostics

Specification or contrast	Estimate	Standard error	Diagnostic p
<i>Panel A. Exact reassignment of Third Front labels among interior provinces</i>			
Baseline age-profile DDD	4.23	1.66	0.059
Predetermined siting controls	2.24	1.15	0.249
<i>Panel B. Nonlinear exposure diagnostics</i>			
Below-median spline slope	2.52	1.35	0.078
Above-median spline slope	2.88	2.73	0.305
Above-minus-below slope	0.36	3.54	0.920
Joint differential-linearity test	$F_{2,20} = 0.41$	—	0.667
Top-minus-bottom exposure quartile	8.99	4.07	0.039
Joint young-age quartile test	$F_{3,20} = 1.73$	—	0.193

Notes. Panel A reports percentage points per common-prefecture standard deviation and conventional province-clustered standard errors. Its p -values enumerate all 715 ways to designate nine of the 13 interior provinces while holding coastal provinces undesignated. Historical designation was not uniform over these assignments, so this is an exact conditional sensitivity diagnostic, not causal randomization inference. Panel B uses the baseline design. Spline slopes are percentage points per 0.10 increase in the heavy-industry share; the knot is the common-sample median. Quartile entries compare the indicated Third Front age-profile contrast with the bottom exposure quartile. The top-quartile coefficient is largest, but the bin pattern is not monotone and its joint test is imprecise.

B.2 Composition outcomes and sex-specific estimates

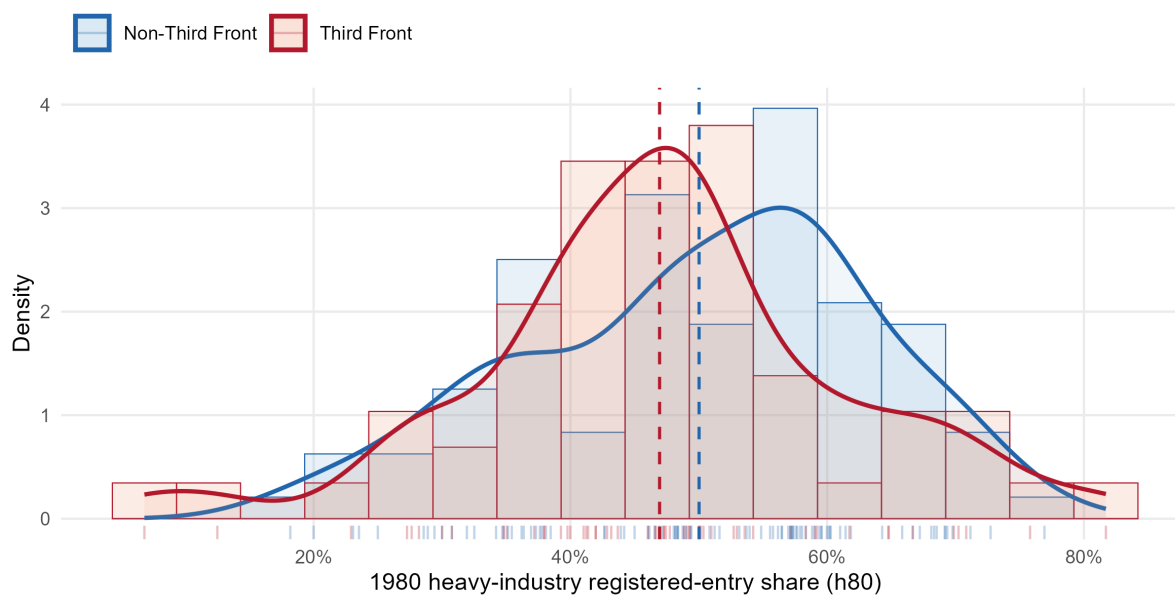
Table B.2. Composition outcomes and sex-specific marriage estimates

	Ages 20–24		Ages 25–29	
<i>Panel A. Composition outcomes, same DDD (1982)</i>				
Currently a student	+0.24	(0.23)	+0.01	(0.04)
Collective dwelling	−0.37	(0.36)	−0.27	(0.21)
Secondary schooling or more	+4.33***	(1.43)	+1.57**	(0.62)
<i>Panel B. Composition outcomes (2000)</i>				
Currently a student	+2.11*	(1.14)	+0.18***	(0.06)
Collective dwelling	+3.31*	(1.88)	+0.52	(0.49)
Secondary schooling or more	+0.59	(1.15)	−0.16	(0.43)
Moved province within five years	+1.65	(1.25)	+0.14	(0.44)
Move reported as marriage-related	−0.42	(0.26)	+0.28	(0.30)
<i>Panel C. Never married, by sex (1982)</i>				
Women	+4.67**	(2.17)	+0.62	(0.54)
Men	+3.79**	(1.56)	+1.55	(0.95)

Notes. Each row applies the core design of Equation (2)—the same support, age contrasts, fixed effects, weights, and province clustering—to the indicated outcome (Panels A–B) or estimates the marriage outcome within sex using province-by-age effects (Panel C). Entries are percentage points per cross-prefecture standard deviation of the 1980 heavy share, contrasted against ages 30–34. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

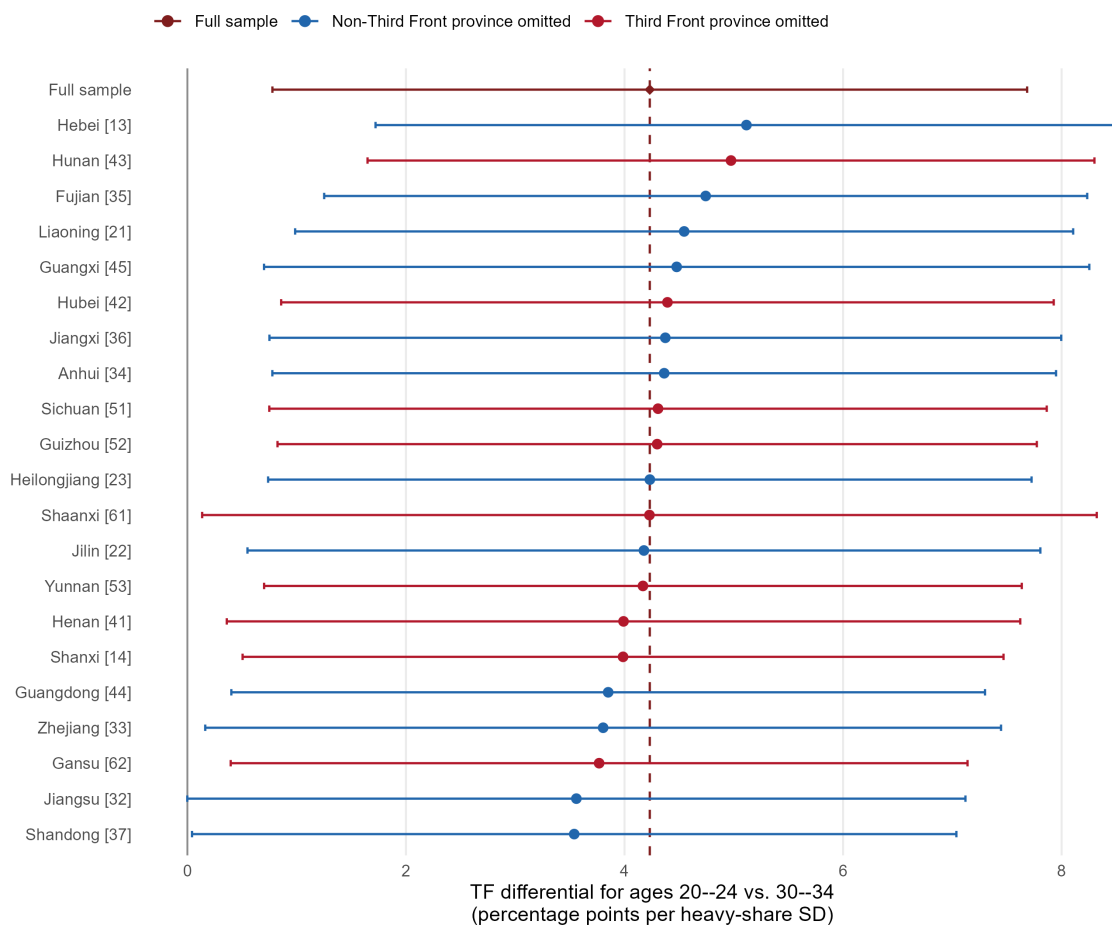
B.3 Exposure support and province influence

Figure B.1. Overlap in the 1980 registered heavy-industry share



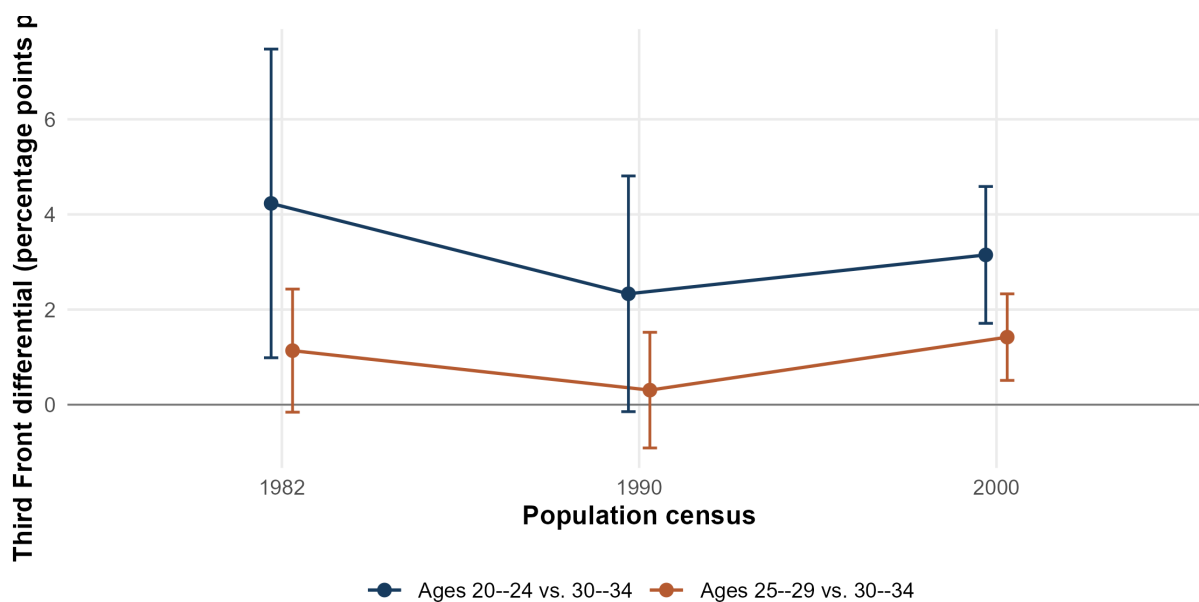
Notes. Unweighted prefecture distributions on the 154-prefecture common geography. Each rug mark is one prefecture; dashed lines are group means. The share is computed from cumulative registered entries, not contemporaneous plant employment.

Figure B.2. Leave-one-province-out influence diagnostic



Notes. Each row re-estimates the 1982 baseline after removing the named province; the full-sample result appears at top and as a dashed vertical line. Intervals use conventional province-clustered standard errors and a $t(G - 1)$ critical value. The exposure scale remains the standard deviation on all 154 common prefectures. This is an influence diagnostic, not a cure for endogenous siting.

Figure B.3. The age-profile differential across censuses



Notes. Estimates reproduce the baseline rows of Table 1. Error bars are conventional 95-percent confidence intervals based on province-clustered standard errors. All three censuses use the same matched prefectures and fixed-effect structure.

C Retrospective and Campaign-Cohort Details

C.1 Empirical objects and populations

Table C.1. Empirical objects, populations, and contrasts

Object	Population	Contrast	Interpretation
1982 resident stock	Residents observed in 1982	Ages 20–24 versus 30–34	Resident-population age gradient
Retrospective profile	Survivors and residents observed in 2000	Reported birth-year or completed-age index	Whether reconstructed cohort histories retain the gradient
Campaign-cohort comparison	Survivors and residents observed in 2000	Marriage risk wholly before or during 1964–1980	Cohort slope shifts under non-overlapping risk windows

Notes. The first two rows address measurement alignment; the third addresses campaign timing. All three differ from a factory-arrival design that assigns exposure using birthplace or pre-marriage residence.

C.2 Full campaign-cohort estimates

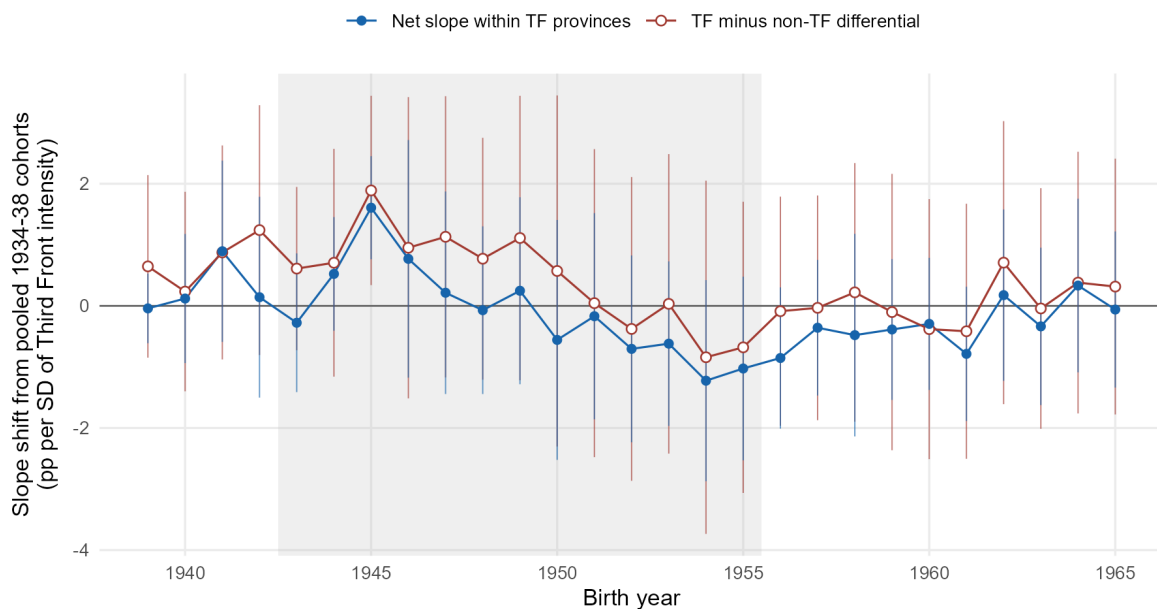
Table C.2. Cohort timing in the 2000 census: broad and calendar-clean comparisons

Threshold	Ordinary non-TF	TF minus non-TF differential			Net shift within TF		
	Estimate (s.e.)	Estimate (s.e.)	Wild p	Holm p	Estimate (s.e.)	Wild p	Holm p
<i>Panel A. Broad cohorts: births 1944–60 versus 1934–43</i>							
Married by 21	−0.16 (0.33)	−0.42 (0.56)	0.515	1.000	−0.57 (0.46)	0.290	1.000
Married by 22	−0.25 (0.64)	−1.03 (0.89)	0.310	1.000	−1.28 (0.62)	0.078	0.475
Married by 23	−0.42 (0.79)	−1.08 (0.97)	0.305	1.000	−1.51 (0.56)	0.039	0.315
Married by 24	−0.31 (0.80)	−0.79 (0.94)	0.444	1.000	−1.11 (0.50)	0.068	0.475
Married by 25	−0.21 (0.67)	−0.28 (0.82)	0.741	1.000	−0.49 (0.47)	0.336	1.000
Married by 26	−0.26 (0.53)	−0.04 (0.66)	0.958	1.000	−0.30 (0.39)	0.462	1.000
Married by 27	−0.14 (0.48)	−0.10 (0.59)	0.865	1.000	−0.24 (0.35)	0.527	1.000
Married by 28	−0.19 (0.45)	+0.02 (0.57)	0.980	1.000	−0.17 (0.34)	0.627	1.000
<i>Panel B. Calendar-clean risk windows: fully within 1964–80 versus fully pre-1964</i>							
By 21 [pre 1926–1942; build 1943–1959]	−0.30 (0.32)	−0.26 (0.50)	0.658	1.000	−0.56 (0.38)	0.223	1.000
By 22 [pre 1926–1941; build 1943–1958]	−0.53 (0.64)	−0.95 (0.84)	0.300	1.000	−1.48 (0.53)	0.015	0.118
By 23 [pre 1926–1940; build 1943–1957]	−0.76 (0.87)	−0.96 (1.03)	0.386	1.000	−1.73 (0.55)	0.027	0.188
By 24 [pre 1926–1939; build 1943–1956]	−0.80 (0.87)	−0.57 (1.10)	0.639	1.000	−1.38 (0.67)	0.090	0.543
By 25 [pre 1926–1938; build 1943–1955]	−0.75 (0.81)	+0.26 (1.07)	0.824	1.000	−0.49 (0.70)	0.517	1.000
By 26 [pre 1926–1937; build 1943–1954]	−0.74 (0.72)	+0.37 (0.92)	0.707	1.000	−0.37 (0.58)	0.583	1.000
By 27 [pre 1926–1936; build 1943–1953]	−0.73 (0.58)	+0.36 (0.77)	0.654	1.000	−0.37 (0.51)	0.499	1.000
By 28 [pre 1926–1935; build 1943–1952]	−0.64 (0.48)	+0.40 (0.59)	0.519	1.000	−0.23 (0.35)	0.533	1.000

Notes. Entries are build-out-minus-pre changes in the heavy-share slope, in percentage points per common-prefecture standard deviation. Ordinary non-TF is the slope change outside Third Front provinces; the differential is the additional change inside Third Front provinces; net within TF is their sum. Parentheses contain conventional standard errors clustered by province. Wild p -values use 4,999 restricted-null wild-cluster draws over 21 provinces; the net null is imposed directly after reparameterizing the primitive coefficients. Holm p -values control the family-wise error rate across the eight thresholds, separately for the differential and net estimands within each panel. Panel A compares births 1944–60 with 1934–43, so threshold-specific marriage calendars can overlap the 1964 and 1980 boundaries. Panel B removes all boundary-straddling risk windows: for threshold a , pre-program births run from 1926 through 1963 − a and build-out births from 1943 through 1980 − a , using the equal-sized cohorts closest to the boundary. This calendar separation comes at the cost of older, more survivor-selected pre-program cohorts. Outcomes are marriage by the indicated age among people unmarried at twenty. Models include prefecture fixed effects and province-by-single-birth-year-by-sex effects, use census person weights, and assign exposure by prefecture of residence in 2000.

C.3 Birth-year timing profile

Figure C.1. Birth-year shifts in marriage by age 25



Notes. Coefficients are changes in the heavy-industry slope relative to pooled 1934–1938 birth cohorts for marriage by age 25 among people unmarried at 20. The figure reports the slope outside designated provinces, the additional Third-Front differential, and the net slope inside designated provinces. Shading classifies each cohort’s ages-21–25 risk window relative to the 1964–1980 build-out. Models include prefecture fixed effects and province-by-single-birth-year-by-sex effects, use population weights, and cluster by province. Industrial composition is assigned by prefecture of residence in 2000.

D Mechanisms and Secondary Designs

D.1 Sex composition among never-married residents

Table D.1. Sex composition among never-married residents

Age range	Differential (SE)	Wild p	Within-TF (SE)	Wild p
Ages 20–34	-1.47 (1.14)	0.235	-2.18 (0.68)	0.052
Ages 20–29	-1.40 (1.11)	0.231	-2.08 (0.69)	0.047

Notes. Outcomes are percentage points of the male share among never-married residents. Differential entries are the heavy-industry-share slope inside Third Front provinces minus the slope elsewhere; within-TF entries are the net slopes inside Third Front provinces. Both are per a one-standard-deviation increase in the 1980 heavy-industry share ($SD = 0.137$). Regressions contain province fixed effects, weight by the number never married, use 148 prefectures in 21 provinces, cluster by province, and use 4,999-draw restricted-null wild inference.

D.2 Direct project measures and the railway IV

Table D.2. Direct Third Front exposure and the 1982 age profile

Treatment	Specification	20–24 vs. 30–34	25–29 vs. 30–34
Third Front investment	Unadjusted	1.61 (1.07)	−0.19 (0.35)
Third Front investment	Siting adjusted	0.46 (1.13)	−0.02 (0.35)
1985 plant employment	Unadjusted	2.34 (1.17)	−0.16 (0.32)
1985 plant employment	Siting adjusted	0.92 (1.37)	0.12 (0.43)
Designated industrial center	Unadjusted	1.20 (2.29)	0.94 (0.58)
Designated industrial center	Siting adjusted	3.93 (2.36)	2.01 (0.76)

Notes. Percentage-point age contrasts per treatment standard deviation, except that the center indicator is a zero-to-one change; conventional province-clustered standard errors are in parentheses. Estimates use 34 Fan–Zou hinterland prefectures in only seven effective provinces, prefecture fixed effects, province-by-single-age-by-sex effects, and population weights. With seven clusters, conventional tail probabilities are unreliable; the table therefore reports no significance stars and is descriptive. Siting-adjusted rows interact 1980 rail distance, mining capacity, terrain, elevation, and distance to the provincial capital with age bins. The direct-investment first stage on 1962 rail distance is weak ($F = 2.39$ under core controls); the plant-capacity first stage is $F = 9.58$.

Given the 22-prefecture overlap, high leverage, and weak-instrument concerns, the wild Anderson–Rubin test is the primary inference; the conventional 2SLS p -value is not.

Table D.3. Plant capacity and the 1982 age profile: railway-IV diagnostic

Age contrast	Specification	OLS	2SLS	First-stage F	Wild AR p
20–24 vs. 30–34	Baseline	1.64 (0.98)	1.98 (1.52)	5.1	0.340
25–29 vs. 30–34	Baseline	−0.01 (0.31)	−0.16 (0.37)	5.2	0.627
20–24 vs. 30–34	Siting controls	1.84** (0.84)	1.96** (0.87)	34.8	0.366
25–29 vs. 30–34	Siting controls	−0.00 (0.24)	−0.18 (0.35)	33.6	0.675

Notes. Coefficients are percentage-point effects of a one-unit increase in the Fan–Zou 1985 non-mining large/medium-plant employment share. The 1962 railway-distance interaction instruments the capacity interaction. All models contain prefecture and province-by-single-age-by-sex fixed effects, weight by population, and cluster by prefecture. Siting controls are 1980 rail distance, mining capacity, slope, elevation, and distance to the provincial capital, each interacted with the age contrast. The overlap contains 22 prefectures. Wild AR p is a 4,999-draw restricted-null wild-cluster test of the reduced form and is robust to weak instruments.

D.3 Work-unit housing in the 1988 CHIP urban sample

Table D.4. Industrial employment and public housing, CHIP 1988 urban sample

	Other household head	Industrial household head	Industrial gap
Non-Third-Front provinces	0.852	0.784	−0.069
Third Front provinces	0.870	0.902	+0.032
Raw difference-in-differences			0.101
Province-FE specification			0.110** (0.036)

Notes. Public housing pools employer-owned and municipal housing-bureau dwellings. The sample contains 8,094 households in ten provinces. The province-fixed-effect model controls for the head’s sex and age and household size and clusters by province. Restricted-null wild-bootstrap $p = 0.0165$; exact designation-reallocation diagnostic $p = 0.0119$. The table establishes differential public-housing dependence, not the effect of a housing queue on marriage.

D.4 CFPS construction and results

The seven CFPS waves from 2010 through 2022 contain 168,307 person-wave records in the access-controlled project extract ([Institute of Social Science Survey, Peking University, 2015](#)), collapsed to one row per person using modal stable characteristics and the earliest valid first-marriage year, birth cohorts 1930–1995, and valid survey city. Controls are sex, ethnicity, and parental education. Respondent education, used only descriptively, is the maximum valid completed years reported across waves. For a threshold a , a person enters if observed to at least a and unmarried at twenty; never-married people carry outcome zero, and people married before twenty are outside the risk set rather than assigned post-event exposure. The fixed-age risk sets contain 16,706 observations at 25 and 16,135 at 28. The regression interacts the city heavy share in the year the person turned twenty with designation, under city and cohort fixed effects.

Table D.5 reports these fixed-age associations.

Table D.5. CFPS fixed-age marriage associations

Exposure construction	Married by 25	Married by 28	Observations (25/28)
Cumulative-entry registry	−1.70** (0.81)	−2.00*** (0.65)	16,706/16,135
Surviving-stock registry	−1.40* (0.79)	−1.50* (0.74)	15,694/15,123

Notes. Entries are percentage-point changes per within-city standard deviation in the heavy-industry share at age twenty, for the Third Front slope relative to the slope elsewhere; city-clustered standard errors are in parentheses. Samples retain respondents unmarried at twenty, including those never married, and require that the threshold age was observed. Models include city and birth-cohort fixed effects plus sex, ethnicity, and parental education. Location is the survey-era city, so these are secondary associations rather than a historical treatment design.

A marriage-year specification—age at first marriage on industrial intensity in the marriage year—is retained only as a diagnostic: it conditions on having married, dates exposure with a year the outcome generates, and gives inconsistent estimates across the two registry constructions on the same respondents; moreover, survey age, marriage year, and age at first marriage satisfy an arithmetic identity within wave, so including survey age with marriage-year effects can mechanically explain the dependent variable. Its canonical interaction is 0.891 years (s.e. 0.455) under city clustering, with province wild-bootstrap $p = 0.096$ and exact interior-designation $p = 0.150$, and falls to 0.401 without city trends.

Matching CFPS cities to the Fan–Zou plant data yields 116 prefectures (21 hinterland). Capacity is associated with 2.70 points less marriage by 25 and 1.54 by 28 across all matched prefectures (wild prefecture-bootstrap $p = 0.060$ and 0.027); hinterland estimates are −2.98 and −5.95 with wild $p = 0.559$ and 0.216. Only three designated centers fall in the usable overlap; their indicator is associated with about five points less marriage at both thresholds, with wild $p = 0.082$ and 0.114. These are descriptive, few-site associations. We report no linked CFPS city names or below-province outputs. The railway IV in this overlap has first-stage F near five and uninformative 2SLS estimates.

E Reproducibility and Data Availability

The analysis was exploratory and not preregistered. The replication package rebuilds the province-safe census crosswalk, constructs the analysis samples from raw or licensed source files, and generates every manuscript exhibit. The complete script-to-exhibit map, validation checks, and data-access restrictions are recorded in `REPLICATION.md`.